

Visualizing and Analysing the Data

Unvarient Analysis

Description

Univariate analysis examines each dataset feature individually to understand its distribution, frequency, and pattern without cross-comparison. This foundational step clarifies how single variables, both numerical and categorical, behave. It reveals normal distributions, skewness, or imbalances, which directly informs better preprocessing and modeling decisions before applying machine learning algorithms.

Distribution plots:

Distribution plots visualize the spread of numerical data to determine if values follow a normal, skewed, or irregular pattern. By displaying where data points are concentrated, these plots identify the central tendency and shape of features like temperature or rainfall. This understanding guides the selection of appropriate data scaling and preprocessing techniques.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load the Excel dataset
```

```
df = pd.read_excel("rainfall in india 1901-2015.xlsx")
```

```
# Display first five records
```

```
print("First 5 records:")
```

```
print(df.head())
```

```
# Distribution Plot for ANNUAL rainfall
```

```
plt.figure(figsize=(8,5))
```

```
sns.histplot(df["ANNUAL"], bins=20, kde=True, color="skyblue")
```

```
plt.title("Distribution Plot of Annual Rainfall")
```

```
plt.xlabel("Annual Rainfall (mm)")
```

```
plt.ylabel("Frequency")
```

```
plt.grid(True, linestyle='--', alpha=0.5)  
plt.show()
```

